

D:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\PR\PID\2000-F00-PID-PR-10001.DWG 290CT24 8:55AM NATASHA BRANCO

A		B		C		D		E		F		G		H		J		K		L		M					
LINE DESIGNATIONS				FLUID SERVICE IDENTIFICATION				MANUAL VALVE NUMBERING				ABBREVIATIONS				GENERAL SYMBOLS				NOTES							
LINE NUMBER UU-VVVV-WWW-XXXX-YYY-ZZ EQUIP. DESCRIPTION PROJ-AREA-DRG-DISC-NUM MEDIUM TO DRAWING NUMBER VALVE SIZE/TYPE IDENTIFIER STATE XXX-YYY-ZZ FROM DRAWING NUMBER				NOTE 4 AA AERATION AIR AE ACID EFFLUENT AS ANTISCALANT BA BLOWER AIR BE BARREN ELUATE BW BORE WATER (HYPERSALINE) CA COMPRESSED AIR CN CYANIDE SOLUTION CO COPPER SULPHATE CON CONCRETE SLURRY COW COOLING WATER CR CRUSHER OIL CS CYANIDE SLURRY CT CALCINE TAILINGS CW CYANIDE WATER DCW DECANT CYANIDE WATER DL DIESEL FUEL FA FLOTATION AIR FD FLOOR DRAINAGE FL FLOCCULANT FR FROTHER FS FLOTATION SLURRY FT FINAL TAILINGS FW FIRE WATER GWH GLAND WATER (HYPERSALINE) HCL HYDROCHLORIC ACID HO HYDRAULIC OIL HP HYDROGEN PEROXIDE HPE HYDROCHLORIC ACID / PREGNANT ELUATE IA INSTRUMENT AIR LN LEAD NITRATE LO LUBRICATING OIL LPG FUEL GAS LS LIME SLURRY MS MILL SLURRY NS CAUSTIC SODA OG OFF GAS OXY OXYGEN PE PREGNANT ELUATE POT POTABLE WATER PSW PROCESS SCHEME WATER PW PROCESS WATER (HYPERSALINE) SC CARBON SLURRY SW SHOWER WATER TE TERIC TH THERMAL OIL TW TRANSFER WATER V VENT WM UFG MEDIA AND PROCESS WATER WW WASHDOWN WATER XA XANTHATE				NOTES 2, 3 VALVE SYMBOL (DEFAULT OPEN) XXXXXXXX LO XXXX UNIQUE IDENTIFIER VALVE SIZE				ATM ATMOSPHERE AG ABOVE GROUND AHR AIR HOSE REEL BDV BLOWDOWN VALVE BL BATTERY LIMIT BT BARRED TEE CA CORROSION ALLOWANCE CC COLD CONSERVE CCTV CLOSED CIRCUIT TELEVISION CONC CONCENTRIC DP DIFFERENTIAL PRESSURE ECC ECCENTRIC ESD EMERGENCY SHUTDOWN ET ELECTRICALLY TRACED FB FULL BORE FC FAIL CLOSED FG FOAM GENERATOR FH FIRE HYDRANT FHR FIRE HOSE REEL FIK FLANGE INSULATION KIT FL FAIL LAST STATE FM FIRE MONITOR FO FAIL OPEN FS FIRE SPRINKLER HDPE HIGH DENSITY POLYETHYLENE HT HEAT TRACE IA INSTRUMENT AIR LC LOCKED CLOSED LO LOCKED OPEN MIT MONOLITHIC INSULATING JOINT NC NORMALLY CLOSED NNF NORMALLY NO FLOW NO NORMALLY OPEN OD OUTSIDE DIAMETER O/F OVERFLOW O/S OVERSIZE PP PERSONAL PROTECTION PSA PRESSURE SWING ADSORPTION PWHT POST WELD HEAT TREATED RO RESTRICTION ORIFICE RS REMOVABLE SPOOL SC SAMPLE CONNECTION SDV SHUTDOWN VALVE ST STEAM TRACED TI THERMAL INSULATION UG UNDERGROUND U/F UNDERFLOW U/S UNDERSIZE WHR WATER HOSE REEL				AG UG ABOVE GROUND / UNDERGROUND BL BY OTHERS PRIMERO BATTERY LIMIT - SCOPE VENDOR PRIMERO BATTERY LIMIT - VENDOR FLOW DIRECTION HOLD DIAMOND NUCLEAR PIPING INSTR. PIPING / INSTRUMENT SPECIFICATION BREAK SLOPE - GENERAL 1:100 SLOPE - SPECIFIED SPEC 'A' SPEC 'B' SPECIFICATION BREAK PIPE MATERIAL TIP XXX TIE-IN POINT				1. P&ID'S SHALL BE READ IN CONJUNCTION WITH: INSTRUMENT LIST LINE LIST MECHANICAL EQUIPMENT LIST PIPE SPECIALS LIST VALVE LISTS 2. REFER TO THE PROJECT SPECIFIC PIPING MATERIALS AND VALVE SPECIFICATION. 3. REFER TO THE PLANT NUMBERING SPECIFICATION 2000-F00-ST5-GE-10003 FOR DETAILS. 4. REFER TO THE PROJECT SPECIFIC PIPING MATERIALS AND VALVE SPECIFICATION FOR PROJECT SPECIFIC FLUID SERVICE IDENTIFIERS. 5. NOMINAL PIPE DIAMETER (DN) FOR STEEL PIPES AND PIPE OUTSIDE DIAMETER (OD) FOR POLYETHYLENE PIPES.							
LINE NUMBER				VALVE TYPE CODE				ACTUATED VALVE NUMBERING				ACTUATED VALVE CODE				PIPING SPECIAL ITEMS				PIPE SPECIAL TYPE				VENDOR ITEMS AND BY OTHERS			
NOTES 2, 3, 5 PIPING UU-VVVV-WWW-XXXX-YYY-ZZ INSULATION (IF REQUIRED) PIPE SIZE MATERIAL FLUID SERVICE UNIQUE IDENTIFIER PROCESS AREA				FLV FLUSHING WATER WATER GLV GLAND WATER VALVE RV RELIEF VALVE SCV SCOUR / DRAIN VALVE TV TECH TAYLOR VALVE VB BALL VALVE VBF BUTTERFLY VALVE VC CHECK VALVE VD DIAPHRAGM VALVE VDT DART VALVE VF FLOAT VALVE VG GATE VALVE VGL GLOBE VALVE VKG KNIFEGATE VALVE VN NEEDLE VALVE VP PRESSURE / SUSTAINING VALVE VPI PINCH VALVE VPL PLUG VALVE VR ROTARY VALVE VTW 3 WAY VALVE				YYY XXXX UNIQUE IDENTIFIER ACTUATOR SYMBOL VALVE SYMBOL XXXXXXXX FC VALVE CODE (OPTIONAL) VALVE FAIL STATE (AS REQUIRED) VALVE SIZE (OPTIONAL)				BDV BLOWDOWN VALVE DCV DENSITY CONTROL VALVE FCV FLOW CONTROL VALVE LCV LEVEL CONTROL VALVE PCV PRESSURE CONTROL VALVE PRV PRESSURE RELIEF VALVE PSV PRESSURE SAFETY VALVE SDV SHUTDOWN VALVE (OPEN/CLOSE) SSV SLAM SHUT VALVE SV SOLENOID VALVE TCV TEMPERATURE CONTROL VALVE TRV THERMAL RELIEF VALVE TSV THERMAL SAFETY VALVE XV ISOLATION VALVE (OPEN/CLOSE)				SP XX-YYY SEQUENTIAL NUMBER PIPE SPECIAL TYPE				EB EXPANSION BELLOWS FR FILTER / REGULATOR (AIR OR GAS SERVICES) FS MISCELLANEOUS FIRE SERVICES ITEMS HS HOSE MC MINSUP COUPLING OR EQUIVALENT MS MISCELLANEOUS ITEMS MV MARIC VALVE OR EQUIVALENT OP ORIFICE PLATE SC STRAUB COUPLING OR EQUIVALENT SP SPRAY NOZZLE SS SPECIAL SPOOL ST STRAINER							
MECHANICAL PIPING				MECHANICAL PLATEWORK				STREAM NUMBERING				PIPE SPECIAL TYPE				VENDOR ITEMS AND BY OTHERS				NOTES							
MECH XXX-MECH-YYYY MATERIAL LINE SIZE				123 STREAM No (UP TO 3 DIGITS)				LINE TYPES EXISTING SERVICE FUTURE SERVICE INTERMITTENT FLOW PRIMARY SERVICE SECONDARY SERVICE OR UTILITY				INSULATION CC COLD CONSERVE ET ELECTRICALLY TRACED INSULATION HT HEAT TRACED INSULATION - GENERAL PP PERSONAL PROTECTION ST STEAM TRACED INSULATION TI THERMAL INSULATION				SP XX-YYY SEQUENTIAL NUMBER PIPE SPECIAL TYPE				EB EXPANSION BELLOWS FR FILTER / REGULATOR (AIR OR GAS SERVICES) FS MISCELLANEOUS FIRE SERVICES ITEMS HS HOSE MC MINSUP COUPLING OR EQUIVALENT MS MISCELLANEOUS ITEMS MV MARIC VALVE OR EQUIVALENT OP ORIFICE PLATE SC STRAUB COUPLING OR EQUIVALENT SP SPRAY NOZZLE SS SPECIAL SPOOL ST STRAINER				1. P&ID'S SHALL BE READ IN CONJUNCTION WITH: INSTRUMENT LIST LINE LIST MECHANICAL EQUIPMENT LIST PIPE SPECIALS LIST VALVE LISTS 2. REFER TO THE PROJECT SPECIFIC PIPING MATERIALS AND VALVE SPECIFICATION. 3. REFER TO THE PLANT NUMBERING SPECIFICATION 2000-F00-ST5-GE-10003 FOR DETAILS. 4. REFER TO THE PROJECT SPECIFIC PIPING MATERIALS AND VALVE SPECIFICATION FOR PROJECT SPECIFIC FLUID SERVICE IDENTIFIERS. 5. NOMINAL PIPE DIAMETER (DN) FOR STEEL PIPES AND PIPE OUTSIDE DIAMETER (OD) FOR POLYETHYLENE PIPES.			
EQUIPMENT NUMBERING				ACTUATED VALVE CODE				PIPE SPECIAL TYPE				VENDOR ITEMS AND BY OTHERS				NOTES											
NOTE 3 AREA EQUIPMENT TYPE EQUIPMENT NUMBER XXX-YY-ZZZ EQUIP NAME EQUIPMENT NAME (AS REQUIRED) EQUIP DESCRIPTION EQUIPMENT DESCRIPTION (AS REQUIRED)				1 29OCT24 LINE TYPES MODIFIED 0 20AUG24 APPROVED FOR CONSTRUCTION AK JB LV LS DM DRW CHECK DESIGN TECH APP PROJ APP				ACTUATED VALVE CODE BDV BLOWDOWN VALVE DCV DENSITY CONTROL VALVE FCV FLOW CONTROL VALVE LCV LEVEL CONTROL VALVE PCV PRESSURE CONTROL VALVE PRV PRESSURE RELIEF VALVE PSV PRESSURE SAFETY VALVE SDV SHUTDOWN VALVE (OPEN/CLOSE) SSV SLAM SHUT VALVE SV SOLENOID VALVE TCV TEMPERATURE CONTROL VALVE TRV THERMAL RELIEF VALVE TSV THERMAL SAFETY VALVE XV ISOLATION VALVE (OPEN/CLOSE)				SP XX-YYY SEQUENTIAL NUMBER PIPE SPECIAL TYPE				EB EXPANSION BELLOWS FR FILTER / REGULATOR (AIR OR GAS SERVICES) FS MISCELLANEOUS FIRE SERVICES ITEMS HS HOSE MC MINSUP COUPLING OR EQUIVALENT MS MISCELLANEOUS ITEMS MV MARIC VALVE OR EQUIVALENT OP ORIFICE PLATE SC STRAUB COUPLING OR EQUIVALENT SP SPRAY NOZZLE SS SPECIAL SPOOL ST STRAINER				1. P&ID'S SHALL BE READ IN CONJUNCTION WITH: INSTRUMENT LIST LINE LIST MECHANICAL EQUIPMENT LIST PIPE SPECIALS LIST VALVE LISTS 2. REFER TO THE PROJECT SPECIFIC PIPING MATERIALS AND VALVE SPECIFICATION. 3. REFER TO THE PLANT NUMBERING SPECIFICATION 2000-F00-ST5-GE-10003 FOR DETAILS. 4. REFER TO THE PROJECT SPECIFIC PIPING MATERIALS AND VALVE SPECIFICATION FOR PROJECT SPECIFIC FLUID SERVICE IDENTIFIERS. 5. NOMINAL PIPE DIAMETER (DN) FOR STEEL PIPES AND PIPE OUTSIDE DIAMETER (OD) FOR POLYETHYLENE PIPES.							
DRG No.		REFERENCES		REV		DATE		DESCRIPTION		DRAW		CHECK		DESIGN		TECH APP		PROJ APP		INFORMATION AND REPRESENTATIONS ON THIS DRAWING ARE THE INTELLECTUAL PROPERTY OF PRIMERO UNLESS SPECIFICALLY IDENTIFIED OTHERWISE. ALL RIGHTS INCLUDING USE RESERVED							
2000-F00-PID-PR-10007		LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM																		PROJECT: NORTHERN STAR RESOURCES LTD KCGM GROWTH PROJECT							
2000-F00-PID-PR-10006		LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM																		TITLE: GENERAL LEGEND SHEET 1 PIPING AND INSTRUMENTATION DIAGRAM							
2000-F00-PID-PR-10005		LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM																		SHEET A1							
2000-F00-PID-PR-10004		LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM																		SCALE NTS							
2000-F00-PID-PR-10003		LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM																		DRG No. 2000-F00-PID-PR-10001							
2000-F00-PID-PR-10002		LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM																		REVISION 1							
2000-F00-PID-PR-10001		LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM																									

